

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st term Regular Examination, 2023
CE 2107
(Construction Technology and Materials)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is brick? What are the field tests of bricks? (10)
(b) Write short notes on: (08)
i) Tempering of bricks; ii) Frog mark; iii) Efflorescence; iv) Jhama bricks
(c) What are the methods available for burning of bricks? Which method is most popular in Bangladesh and describe it. (17)
2. (a) What are the benefits of cement over lime? Why lime is not used for concrete? (10)
(b) What is setting time for cement? What is initial set, final set, false set and flash set? (10)
(c) The volume of moist sand was found to be 100 inch³ and the volume of the saturated sand is 85 inch³. What is the %bulking and bulking factor of the sand? (05)
(d) What is FRP? Briefly explain its benefits. (10)
3. (a) What is mortar? What are the types of mortars? (08)
(b) What are the properties of good mortar and why mortar is used in structures? (11)
(c) What are the field tests of cement? (10)
(d) Write short notes on: i) Segregation of concrete; ii) Bleeding of concrete; iii) Creep of concrete (06)
4. (a) What are the uses of sand in concrete? (07)
(b) Draw the qualitative diagram of gradation curve for well graded, gap graded and uniformly graded sample. (09)
(c) Sieve analysis of two varieties of sands are shown in the table. The sample size is 500 gm. Calculate the ratio of each sand to find out the fineness modulus of (19)

Sample size = 500 gm

Sieve Size	Weight Retained (gm)	
	Sample A	Sample B
#4	0% of sample size	0% of sample size
#8	6.5% of sample size	0% of sample size
#10	15% of sample size	0% of sample size
#16	15% of sample size	0% of sample size
#20	30% of sample size	6% of sample size
#30	10% of sample size	5% of sample size
#50	20% of sample size	50% of sample size
#100	2% of sample size	38% of sample size
Pan	1.5% of sample size	1% of sample size

Section – B

5. (a) Define foundation. Discuss various types of shallow and deep foundation with neat sketch. (15)
- (b) What are the problems and precautions for the safety of foundations in black cotton soil? (10)
- (c) A drop hammer with dimension 3'X3'X0.5' is made by steel falling from a height of 20 inch results in an average penetration of 2 inch in the first few blows. The weight of the pile resting on its head maybe taken as 1 kip. [$\gamma_{steel} = 490 \text{ lb/ft}^3$] Determine the ultimate load bearing capacity of the pile. (10)
6. (a) Briefly describe various methods for dealing with foundations on loose soil with neat sketch. (12)
- (b) Draw a typical plan and section of a steel grillage foundation. (08)
- (c) Define the following terms with neat sketch: (06)
- i) King closure
 - ii) Three quarter bat
 - iii) Heading bond
- (d) Write down the comparative merits and demerits of English and Flemish bond. (09)
7. (a) What are the objectives of plastering? Describe the defects found in plaster. (10)
- (b) What are the main differences between floor and roof? Write down the factors to be considered for the selection of suitable floor types. (12)
- (c) What is stair? What are the types of stair? Also describe the requirements of a good stair. (13)
8. (a) Draw a typical section of reinforced concrete lintel with chajja. (05)
- (b) Is it possible to plaster over paint? Give proper explanation. (05)
- (c) Write down the position of cavity wall and discuss about constructional details. (13)
- (d) Write the names of different tools and equipment that used in road construction. (12)
- Describe mass haul diagram with neat sketch.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Examination, 2023
URP 2191
(Statistics For Planners - I)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary charts and graphs will be provided (Normal distribution table).
iv) Assume data where necessary.

Section – A

1. (a) "Statistics relies heavily on how the numbers are chosen and how the statistics are interpreted"- Explain the statement with an example. (05)
- (b) Differentiate between descriptive and inferential statistics with example. (05)
- (c) Draw a good table and explain the major properties in short. (07)
- (d) Draw histogram, frequency polygon and ogive curve from the following data set. (18)

Class	0-<10	10-<20	20-<30	30-<40	40-<50	50-<60	60-70
Frequency	3	7	13	17	14	9	2

2. (a) What are the advantages and disadvantages of Arithmetic mean? Write short notes on alternatives for arithmetic mean with example. (10)
- (b) Calculate mean, median and mode of the following data. Can you make an educated guess about the normality of the data? (25)

Class	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Frequency	4	8	17	22	19	12	5

3. (a) Prove that the first central moment is always zero and second central moment is variance. (10)
- (b) Calculate (i) Q1(1st quartile) (ii) D8 (8th-decile) and (iii) P75 (75th percentile) from the following data: - (15)

Class	Frequency
13-15	12
15-17	14
17-19	50
19-21	60
21-23	32
23-25	20
25-27	10

- (c) The worker's age is given below: - 21,26,29,31,34,39. Compute Kurtosis and also comment on the nature of the distribution. (10)
4. (a) Write short notes on the following: (35)
 - i. Types of statistical inquiry
 - ii. Key points of framing a questionnaire
 - iii. Geometric mean
 - iv. Box-whiskers plot
 - v. Positively skewed Vs Negatively skewed

Section – B

5. (a) Explain five basic words in Statistics with appropriate examples. (10)
- (b) When a Project is failed, there is a 75% chance that it is due to corruption and a 15% chance that it is due to a budget problem. There is an 85% chance that it is due to a corruption or a budget problem. What is the probability that both of these problems are at fault? (10)
- (c) Explain different levels of measurement of statistical data with appropriate examples. (15)
6. (a) Define with example: (15)
- (i) Sample space; (ii) Event; (iii) Dependent and independent event; (iv) Mutually exclusive event; (v) Conditional probability; (vi) Exhaustive event (vii) Simple event
- (b) A household owner is currently considering insuring the contents of his house against theft for one year. He estimates that the contents of his house would cost him \$17,000 to replace. Local crime statistics indicate that there is a probability of 0.3 that his house will be broken into in the coming year. Now there are 4 scenarios for him. (20)
- i. His losses would be 10%, 20%, or 40% of the contents with probabilities 0.5, 0.35 and 0.15 respectively with no policy insurance.
- ii. An insurance policy from company A costs \$150 and in that event his losses would be 17%, 27%, or 47% of the contents with probabilities 0.4, 0.45 and 0.15 respectively.
- iii. Company B costs cheaper at \$100 a year but losses would be 30%, 40%, or 50% of the contents with probabilities 0.7, 0.15 and 0.05 respectively.
- iv. Company C costs even cheaper at \$75 a year but losses would be 18%, 21%, or 15% of the contents with probabilities 0.15, 0.15 and 0.7 respectively.
- Draw the decision tree and make Decision with proper explanation.
7. (a) Explain Sampling Bias with example? (05)
- (b) Briefly describe Principles and Methods of Sampling? (10)
- (c) New Delhi Railway Station has a single ticket counter. During the rush hours, customers arrive at the rate of 17 per hour. Find out the Probability that there are 5 customers in 30 Minutes. (10)
- (d) If it is more than 40° the owner of the Juice shop makes \$17 per hour. If it is less than 25°, he can lose \$5 per hour. What is his expectation if the probability of more than 40° is 0.8. (10)
8. (a) Describe how standardization works in a normal probability distribution. Why is standardization required? (10)
- (b) Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination. Scores on the GMAT are roughly normally distributed with a mean of 527 and a standard deviation of 112. What is the probability of an individual scoring above 500 on the GMAT? How high must an individual score on the GMAT in order to score in the highest 17%? (15)
- (c) Briefly elaborate the characteristics of Normal Probability Distribution's. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2023

URP2115
(Urban Planning Techniques)

Full Marks: 210

Time: 3.00hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) How urban development plan is essential for achieving balanced growth and enabling all areas to develop to their maximum potential? Explain with proper examples. (10)
(b) "Urban development plan integrates various components of an urban area"- provide a brief overview of those components. (10)
(c) Draw a hierarchical order using different forms of urban development plan. Are there any drawbacks of maintaining the hierarchy? Explain with examples in the context of Bangladesh. (15)
2. (a) "Master plan is necessary to give a perspective picture of a fully developed town". Illustrate with appropriate examples. (12)
(b) Identify the challenges that hinder the implementation of master plan in practical scenarios with relevant examples. (15)
(c) What aspects of the Detailed Area Plan contributes to resolve the issues found in the Master Plan? (08)
3. (a) "The four criteria of Structure Plan: degree of detail, level of interaction, time duration of interaction & scale of effect help determining the nature of decision making". Explain with appropriate examples. (10)
(b) How does strategic Plan guide the policy formulation and strategic decisions of the development of an urban area, aligning with national development goals? (10)
(c) Make a comprehensive analysis among structure Plan, Master Plan and Detailed Area plan. (15)
4. (a) Critically assess the Land Consolidation technique of participatory area planning. (10)
(c) Make a comparative analysis between Guided Land Development Plan and land Readjustment Plan. Which one is more feasible for building a new residential area regarding the socio-economic condition of Bangladesh? (15)
(c) "Land Banking is a key policy instrument of government for spontaneous development and regulating land prices"- explain with necessary examples. (10)

Section – B

5. (a) Explain the following: (15)
 - I. SIC
 - II. Basic and non-basic sector
 - III. Base or Income multiplier
- (b) Distinguish between population Estimate, Projection and Forecast. "Population projections are often mislabeled as forecast" – explain. (10)
- (c) Explain population cohorts and pyramids with the help of an appropriate diagram. (10)

6. (a) What are the assumptions and problems of LQ approach? How is minimum requirement approach different and is it superior? (15)
- (d) Using LQ approach, find out the basic and non-basic sectors of the following economic sectors. (10)

SIC code	Industry	Local	National
A	Agricultural	5140	70,96,533
C	Construction	2379	4642735
D	Food production	2688	16,014
I	Hotels	178	1,33,4210
H	Government	4117	9,32,6667
Total		12,489	22,41,6159

- (c) Given the following: (10)
- Projected growth rate for industry i in a region from year t to t" = 15%
- Local employment in industry i in year t = 2600
- Projected local employment shift for industry i for the period t to t" = 12.5%
- Calculate projected local employment in industry i for year t" using constant share approach and shift share approach

7. (a) Calculate the total new housing units required for a population of 65000. (15)

Assumption	Value
Group Home	
Proportion of persons living in Group Homes	4%
Housing Mix	
Single Family	70%
Multi-Family	30%
Household Size	
Single Family	2.5
Multi-Family	1.5
Vacancy Rate	
Single Family	5%
Multi-Family	13%

- (b) Describe the values and functions of open space. (10)
- (c) Briefly write about the Triangulation assessment technique. Provide example of Quantitative technique. (10)
8. (a) Explain Suitability Analysis with the help of a proper example demonstrating the calculations that take place. (15)
- (b) Depict the Level of Intervention in Urban Renewal with the help of a diagram (10)
- (c) Explain the Build Out Analysis methods with the help of a proper example (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year Regular Examination, 2023
Hum 2117
(Sociology for Planners)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section A

1. (a) How is sociology related to urban planning? Analyze. (10)
(b) Find out the relationship between society and production based on Karl Marx's mode of society. (10)
(c) Describe how technological development has shaped the history of human societies. (15)
2. (a) What is ethnocentrism and what are the disadvantages of it? (10)
(b) Analyze biological and cultural perspectives of human behavior. (15)
(c) Why cultural lag has become a problem in modern society? (10)
3. (a) What are the social foundations of deviance? (10)
(b) What are the problems with social conformity? (10)
(c) Explain control theory developed by Travis Hirschi to juvenile delinquency. (15)
4. (a) What is Exurb? (05)
(b) How does Louis Wirth describe urbanism? (15)
(c) What is urban ecology? Illustrate spatial distribution of urban groups. (15)

Section – B

5. (a) Write the basic characteristics of urban social problem. (05)
(b) Define crime and juvenile delinquency. Describe the most effective way to prevent both the harmful activities. (15)
(c) As an urban planner, how do you make an acceptable rehabilitation plan for the vagrant people of urban areas? (15)
6. (a) Define kinship. Discuss the impacts of it on the social life of Bangladesh. (15)
(b) Explain marriage and family. Do you think both have huge significance in the globalized society for social stability? Show your logic with appropriate examples. (20)
7. (a) What is leadership? Explain the leadership pattern theoretically. (10)
(b) What is social control and what are the agencies of it? (15)
(c) Do you think that the traditional role of the agencies of social control is changing rapidly in the modern society? Analyze. (10)
8. (a) What is prostitution? Discuss the impact of it on young people. (15)
(b) What are the key factors for characterizing Bangladeshi people as unique in the world population? (10)
(c) Religion influences our entire social life. Give your opinion by showing practical experiences. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2023
URP 2113
(Site and Area Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) Can urban rural continuum be responsible for SMART Growth? Briefly discuss with example (15)
- (b) A highway has a design speed of 70 mph and a super elevation rate of 0.01. If $f_s = 0.15$, what should be the radius of the curve? (10)
- (c) What is warm humid zone? Briefly discuss about the orientation of the building and the room arrangement within building in warm humid zone (10)
- 2 (a) "The distinction between active and passive land use regulation is nonexistent"- do you agree? Give your answer for or against this statement. (15)
- (b) What are the climatic elements in the site planning and how one can modify their presence? (10)
- (c) One of three sites to be selected for a plant and it is found that 7 factors influence the decision of location. These factors had been weighted. Find out which site is best to be selected? (10)

Location Factor	Weight	Site A	Site B	Site C
Labor	0.30	80	65	90
Transportation	0.20	100	91	75
Utilities	0.15	60	95	72
Construction	0.15	75	80	80
Taxes	0.10	65	90	95
Waste disposal	0.05	85	92	65
Miscellaneous	0.05	50	65	90

- 3 (a) Why zoning is necessary for land use control? Explain the following land use control terms: (15)
i) Defeasible fees, ii) Eminent domain, iii) Easements, iv) Equitable servitudes; (v) Restrictive covenants
- (b) Write down the functions of open spaces. (10)
- (c) Prepare a wind rose diagram following the data in the table (10)

Wind Direction	Percentage of Days Wind Blowing from the Direction
North	27
North East	9
East	8
South East	14
South	10
South West	7
West	6
North West	15
Calm	4

- 4 (a) Explain the four basic models of site planning with proper diagram. (15)
- (b) Briefly explain the need of land use control with necessary examples. (10)
- (c) "State land use controls are passive land use controls"- give your answer for or against this statement. (10)

Section – B

- 5 You are tasked with designing a residential neighborhood that spans 200 acres. This project involves creating residential plots that are 3, 4, and 5 kathas in size. Given a target population density of 250 people per acre, answer the following questions of the project:
- (a) Explain your understanding of neighborhood planning. How are different blocks designated within a neighborhood? (12)
 - (b) Calculate the number of primary and secondary schools required to serve the anticipated population, following international standards. (08)
 - (c) Calculate the dimensions (length and width) of the 3, 4, and 5 katha plots, adhering to a golden rectangle ratio of 1:1.618. (15)
- 6 Neighborhoods are integral components of cities and also a part of the country and world. The adage 'Think globally, act locally' underscores contemporary approaches to development. As a Planner, it's essential to integrate various technologies and methodologies during the neighborhood planning process.
Read the aforementioned statement and answer the following questions.
- (a) What technologies are currently available for subdivision planning and from your perspective, which one is the most effective? (10)
 - (b) List ten key principles that should guide neighborhood planning. (10)
 - (c) Neighborhood should be planned keeping the priority of the children in mind. Write down some suggestions for child-friendly neighborhood development as a planner. (15)
- 7
- (a) Briefly explain the model of site planning process. (10)
 - (b) Write down the steps of preparing a Shadow Diagram of a site. (15)
 - (c) You are an urban planner tasked with evaluating the impact of a new 10-meter-tall building on its neighboring park for solar access. You need to calculate the length of the shadow cast by this building at noon on the winter solstice. Assume you are located at a latitude where the sun's elevation angle (the angle above the horizon) at noon on the winter solstice is 30 degrees. (10)
- 8
- (a) Why lot design is a critical component of Site Planning? Write down any three principles of lot design in site planning with proper examples. (15)
 - (b) Write a short note on the following terms. (20)
 - Curvilinear pattern
 - Cul-de-sac pattern
 - Cluster pattern
 - Rectilinear pattern